

Tutorial Sheet - Topic 08

Programming Fundamentals 1

Objects, Classes, and Arrays , Boolean Methods

This tutorial reinforces your understanding of **classes**, **objects**, **arrays of objects**, and **driver programs**. Work through each question on paper.

Exercise 1 – Object-Oriented Concepts

Fill in each blank with the correct term:

1. A _____ defines a blueprint for creating objects.
 2. An _____ is a single instance of a class.
 3. The keyword used to create a new object is _____.
 4. Encapsulation hides data by making fields _____.
 5. A class groups together both _____ and _____.
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Exercise 2 – Basic Class Definition (Product Class)

Write a class called `Product` that contains:

- `String name`
- `double price`
- `boolean inStock`

Tasks:

1. Write a **constructor** that sets all three fields.
 2. Write **getter** and **setter** methods for each field.
 3. Write a **toString()** method returning one formatted line of text.
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Exercise 3 – Writing a boolean method - 1

Write a boolean method called **isValidPrice(double price)** that returns true if the price is between 0 and 100 and false otherwise.

Exercise 4 – Writing a boolean method - 2

Write a more general boolean method called **inValidRange(double valToCheck, double min, double max)** that returns true if the price is between min and max and false otherwise.

Exercise 5 – Adding Validation

Modify the `Product` class so that:

- `price` should be between 0 and 100
- `name` must be non-empty and no longer than 20 characters.

Change the setters and constructors to enforce these rules.

Try to use the method you wrote in Exercise 3 and then the method you wrote in Exercise 4.

Exercise 6 – Changing Validation

We now change the validation rules.

Modify the `Product` class so that:

- `price` must be between 1000 and 5000
- `name` must be non-empty and no longer than 30 characters.

Change the setters and constructors to enforce these rules, again using the method you wrote in Exercise 4.

Exercise 7 – GymMember Class (Updated Validation Rules)

Create a `GymMember` class with:

- `String name`
- `double height`
- `double weight`
- `boolean activeMembership`

Tasks:

1. Add a constructor and full set of accessors/mutators.
 2. Apply these **validation rules**:
 - `height` must be between **1.2 and 2.3 metres**.
 - `weight` must be between **35 and 250 kilograms**.
 - `name` cannot be blank and must be at least **3 characters long**. (use the method you wrote in Exercise 4)
 3. Add a `toString()` method that prints all details neatly formatted.
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Exercise 8 – Arrays of Objects (Custom Inventory)

Design a new class `Inventory` that stores multiple `Product` objects in an array.

Tasks:

1. Create a field `private Product[] stock` and initialise it in the constructor.
 2. Write a method `findItem(String name)` that returns the matching product or `null` if no match is found.
 3. Write a method `addItem(Product p)` that inserts a product only if its name is unique and the array is not full.
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Exercise 9 – Extended Functionality

Extend your `Inventory` class to include:

- `countInStock()` → returns how many products have `inStock` set to true.
 - `percentageInStock()` → returns the percentage of total products currently available.
 - `listProducts()` → returns a string listing the names of all products in stock.
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Exercise 10 – Application Integration

Write a simple `main()` method that:

1. Creates an `Inventory` with space for 3 products.
 2. Adds 3 products to the inventory.
 3. Prints the **count in stock**, and the **percentage available**.
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